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SUBJECT CODE NO: - RNP-8014
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group B) (NEP) Sem - I
Examination May/June-2026
Basics of Electrical Engineering

[Time: 3:00 Hours]**[Max. Marks: 60]**

N. B

Please check whether you have got the right question paper.

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Question No. 2, 3, 4, 5, 6 and 7.

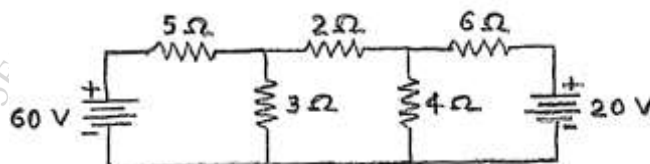
SECTION A**Q1 Solve any 10 from the following questions.****20**

1. The algebraic sum of current at any junction in a network is equal to
 A. Infinity B. One C. Zero D. High
2. The unit of this electrical Power will
 A. Ampere B. Volt C. watt D. Newton
3. Usually, the field is stationery and conductors are rotating in
 A. transformer B. Generator C. Both D. None
4. E.m.f. induced in one coil by the influence of the other coil is called.....
 A. Mutual EMF B. Self EMF C. Both D. EMF
5. The value of Peak Factor of sinusoidal waveform is
 A. 1.11 B. 1.21 C. 0.70 D. 0.63
6. RMS value of sinusoidal wave form is Maximum value.
 A. 1.11 B. 1.21 C. 0.70 D. 0.63
7. Line voltage in Delta connected three phases Phase voltage.
 A. 3-time B. $\sqrt{3}$ C. Equal D. 2 Times
8. The ratio of maximum value to the r.m.s. value of an alternating quantity is known.....
 A. Zero B. Peak Factor C. Form Factor D. None
9. A Practical constant Current source should have internal resistance
 A. Zero B. Not Infinite C. High D. Infinite
10. An ideal current source is that which has an internal resistance
 A. Zero B. Low C. High D. Infinite
11. The phenomenon of lagging of flux density (B) behind the magnetizing force (H) in a magnetic material
 A. Flux B. Current C. B-H Curve D. Density

12. The ability of an inductor store energy in form of
 A. Charge B. Current C. Capacitance D. Voltage
13. An ideal Voltage source is that which has an internal resistance
 A. Infinity B. One C. Zero D. High

SECTION B

- Q.2** a) Define Peak value, RMS value, Form Factor, Peak factor, Average value. **05**
 b) Explain the R-L Series A.C. Circuit. **05**
- Q.3** a) Explain Voltages and Currents in Balanced Y-Connected Supply System with phasor diagram. **05**
 b) Explain Voltages and Currents in Balanced Delta Connected Supply System with phasor diagram. **05**
- Q.4** a) Comparison between Magnetic and Electric Circuits. **05**
 b) A solenoid have 500 turns supply with 20 volts of resistance is 4-ohm, length of solenoid is 12 cm with area is 4 sq.cm, relative permeability is 600 find 1) MMF **05**
 2) Magnetic field intensity 3) Flux density 4) Reluctance
- Q.5** a) Prove the Capacitors in Parallel. **05**
 b) Energy Stored in A Capacitor. **05**
- Q.6** a) Explain Ideal & Practical sources in detail. **05**
 b) Prove the Series and parallel combination of resistor with circuit diagram. **05**
- Q.7** a) Find loop Current i_1 , i_2 & i_3 By Mesh Analysis. **05**



- b) Explain nodal analysis with example. **05**